

EXAMIQ

Functional Testing and Test Cases

A Self-Regulated Mathematics Exam Review System for College Students with Dynamic Feedback, Mistake Tracking, and Confidence-Based Test Flow

1. Scope

This document defines functional test cases (Appendix A) and a functional testing checklist (Appendix B) for ExamiQ. Four user roles are covered: Student, Faculty, Chairperson, and Campus Administrator. Functional testing is limited to five primary functions per role. S = Satisfactory; U = Unsatisfactory.

Role	Post-login home
Student	/student/dashboard/
Faculty	/professor/courses/
Chairperson	/chairperson/dashboard/
Campus Administrator	/admin/ or /campus/

Appendix B — Functional Testing Checklist

Function | Satisfactory (S) / Unsatisfactory (U) | Note

B.1 Student

Function	S/U	Note
Create Account	S	Working — sign up with program, year level, and section
Login Account	S	Working — redirect to /student/dashboard/
Start Review Session	S	Working — setup wizard, timed delivery, session summary
View Mistakes	S	Working — auto-logged wrong answers; list and weak-area patterns
View Dashboard	S	Working — performance summary, session history, topic progress

B.2 Faculty (Professor)

Function	S/U	Note
Create Account	S	Working — creates pending account until campus approval
Login Account	S	Working — redirect to /professor/courses/ after approval
Create Question	S	Working — saves as pending; requires chair approval for live use
Configure Exam Setup	S	Working — enables student review eligibility for section
View Course Analytics	S	Working — roster, heatmap, and intervention data

B.3 Chairperson

Function	S/U	Note
Create Account	S	Working — requires department; pending until campus approval
Login Account	S	Working — redirect to /chairperson/dashboard/
Create Teaching Assignment	S	Working — links faculty, section, subject, and term
Approve Question	U	Service exists but review UI not wired to URLs/views
View Programs Dashboard	S	Working — department program performance and cross-program analytics

B.4 Campus Administrator

Function	S/U	Note
Login Account	S	Working — access to /campus/ or /admin/
Approve / Reject Registration	S	Working — activate or reject pending faculty and chairperson accounts
Create Faculty / Chair Account	S	Working — provision active approved accounts directly
Manage Academic Calendar	S	Working — set current year and term for review eligibility
Manage Program Sections	S	Working — create sections for student signup and assignments

Appendix A — Test Cases

Action | Activities | System Response | Actual Errors | Response

A.1 Student

Action	Activities	System Response	Actual Errors	Response
Sign In	Enter email and password; click Sign In	User authenticated; redirect to /student/dashboard/	Invalid email or password	Prompt to enter valid credentials
Sign Up	Enter name, email, student number, program, year level, section, phone, password	Account created; redirect to student dashboard	Email already registered	Prompt for unique email
Sign Up	Enter weak password	—	Password too weak	Prompt to use a strong password
Sign Up	Select full section	—	Section full	Prompt: This section is full. Choose another section.
View Profile	Update program, year level, or section at /profile/	Profile saved	Section mismatch	Display validation error
Log Out	Click Sign Out	Session ended; redirect to login	No system errors	No additional response
Review Setup	Select subject, topic, difficulty, duration at /student/review/setup/	Wizard advances; preview shown	Profile incomplete	Display eligibility message; block session
Start Review	Confirm setup	Timed session begins at /student/review//	No approved questions	Display error or empty session message
Submit Answer	Answer within time limit	Answer graded; confidence inferred; next question loaded	Time expired	Mark unanswered; advance per timer
Session Summary	Complete session; view summary	Accuracy and calibration insights displayed	No system errors	Summary shown for partial sessions
View Feedback	Open feedback from session summary	Step-by-step explanation displayed	Feedback not generated	Prompt to generate feedback
View Mistakes	Navigate to /student/mistakes/	Mistake records listed	No mistakes yet	Show empty state
View Dashboard	Navigate to /student/dashboard/	Performance summary displayed	No session history	Show zero-state metrics

A.2 Faculty (Professor)

Action	Activities	System Response	Actual Errors	Response
Sign Up	Enter email, employee ID, phone, password; select Faculty	Pending inactive account created	Employee ID missing	Prompt: Employee ID is required.
Sign In	Enter credentials before approval	—	Account pending	Redirect with pending banner
Sign In	Enter credentials after approval	Redirect to /professor/courses/	Invalid credentials	Prompt valid credentials
Log Out	Click Sign Out	Session ended	No system errors	No additional response
Create Question	Enter stem, choices, answer at questions/create/	Question saved as pending	Missing fields	Prompt complete required fields
Edit Question	Edit approved question	Updated; status returns to pending	Not owned	Return 404
Exam Setup	Enable setup, set topics, timer at exam-setup/	Students become eligible	Missing config	Display validation errors
Edit Feedback	Edit explanation steps at feedback/	Steps saved	Question not in course	Return 404
Course Analytics	Open /professor/courses//	Performance summary displayed	Course not owned	Return 404
Roster	Open roster; drill into student	Student performance shown	Student not in roster	Return 404

A.3 Chairperson

Action	Activities	System Response	Actual Errors	Response
Sign Up	Enter email, employee ID, department, password	Pending account created	Department missing	Prompt: Department is required for chairpersons.
Sign In	Enter credentials after approval	Redirect to /chairperson/dashboard/	Invalid credentials	Prompt valid credentials
Log Out	Click Sign Out	Session ended	No system errors	No additional response
Review Questions	Approve or reject pending question	Status updated to approved/rejected	Outside department	Access denied or 404
Review Questions	Access review UI	—	UI not wired	Known gap: use Django admin workaround
Create Assignment	Select faculty, section, subject, term	Assignment and course created	Missing fields	Display validation error
Programs Dashboard	Open /chairperson/dashboard/	Program performance displayed	No data	Show zero-state dashboard
Cross-Program Analytics	Open /chairperson/analytics/by-program/	Analytics by degree program	No student data	Show empty analytics
Course Audit	Open /chairperson/courses/	Course participation listed	No courses	Show empty list

A.4 Campus Administrator

Action	Activities	System Response	Actual Errors	Response
Sign In	Enter admin credentials	Redirect to /campus/ or /admin/	Invalid credentials	Prompt valid credentials
Approve Registration	Approve pending user at /campus/users/	User activated and approved	Not pending	Error: Only pending registrations can be approved.
Reject Registration	Reject pending user	User rejected and inactive	Not pending	Error on invalid state
Create Professor	Enter email, name, employee ID, password	Active professor created	Duplicate email	Prompt unique email
Toggle Active	Toggle approved user active status	is_active flipped	Own account	Error: cannot deactivate own account
Academic Calendar	Set current year and term	Current term saved	Invalid dates	Display validation error
Program Sections	Create section with program and capacity	Section available for signup	Duplicate code	Display validation error
Log Out	Click Sign Out	Session ended	No system errors	No additional response

Appendix C — Cross-Role Access Matrix

#	Actor	URL / Resource	Expected result
1	Anonymous	/student/dashboard/	302 → login
2	Student	/student/dashboard/	200
3	Student	/professor/courses/	403
4	Faculty	/professor/courses/	200
5	Faculty	/student/review/setup/	403
6	Chairperson	/chairperson/dashboard/	200
7	Chairperson	/campus/	403
8	Campus Admin	/campus/	200
9	Campus Admin	/student/dashboard/	403
10	Pending Faculty	/accounts/login/ (valid password)	302 → ?registered=pending

Known Limitations

Chairperson Question Review UI is marked U — templates exist but views/URLs are not wired. Questions may remain pending unless approved via Django admin.